

Bit	Code	Feature	Conformance	Summary
5	FA	ForecastAdjustment	O	Allows an EMS to adjust an ESA's planned operation.
6	CON	Constraint-BasedAdjustment	O	Allows an EMS to request constraints to an ESA's planned operation.

The conformance rules specified above can be described as:

- At least one of the features SHALL be supported.
- At most one of the 'SFR' and 'PFR' features SHALL be supported.
- If one or more of the 'STA', 'PAU', 'FA', or 'CON' features are supported, then either the 'PFR' or the 'SFR' feature SHALL also be supported.
- If PA is supported, SFR SHALL NOT be supported

9.2.4.1. PowerAdjustment Feature

For Energy Smart Appliances (ESA) the definition of being 'smart' mandates that they can report their current power adjustment capability and have an EMS request a temporary adjustment. This may typically be to curtail power requirements during peak periods, but can also be used to turn on an ESA if there is excess renewable or local generation (Solar PV).

For example, a home may have solar PV which often produces more power than the home requires, resulting in the excess power flowing into the grid. This excess power naturally fluctuates when clouds pass overhead and other loads in the home are switched on and off.

EVSE Example: An EMS may therefore be able to turn on the EVSE (if the vehicle is plugged in) and can start charging the vehicle, and periodically modify the charging power depending on PV generation and other home loads, so as to minimize import and export to the grid. An EMS may also use this feature to control the discharging (and re-charging) of the vehicle if the EVSE and vehicle support the V2X feature of the EVSE cluster of the associated EVSE device.

9.2.4.2. PowerForecastReporting Feature

For Energy Smart Appliances (ESA) the definition of being 'smart' implies that they can report their indicative forecast power demands or generation, to a greater or lesser extent. For some ESAs this is highly predictable (in terms of both power and time), in other appliances this is more challenging and only a basic level of forecast is possible.

Forecasts are defined from a current time, using a slot format, where the slot is akin to a relatively constant operating mode.

Washing machine example: a washing machine may have stages of a washing cycle: heating, tum-